



Luis Lopez

luislopezdiaz@cmail.carleton.ca

carleton.ca/APEX

Department of Mechanical and Aerospace Engineering
Ottawa, Canada

Fall 2026

MAAE 2300: Fluid Mechanics I

PA Introduction

Welcome to MAAE 2300 PA Tutorials!

☰ Today's Session Plan

- 1 **Meet Your TA:** Background and contact channels.
- 2 **Icebreaker Dynamic:** A quick, informal round to get to know each other.
- 3 **Tutorial Philosophy:** The 6-step problem-solving framework.
- 4 **Keys to Success:** Practical advice for assignments and exams.

🤝 My Commitment to You

- Tutorials are **interactive workshops**, not passive lectures.
- Safe learning space: questions and mistakes are essential for growth.
- Focus on developing **physical intuition** rather than rote formula memorization.

Meet Your TA

- 1 Meet Your TA
- 2 Icebreaker Activity

- 3 Tutorial Methodology
- 4 Keys to Success

About Your Teaching Assistant

Information

Luis López

PhD Candidate

› Research Focus

- ›› TSO–DSO coordination and smart grids
- ›› Machine learning for grid flexibility & DERs

› Technical Toolkit

- ›› Mathematical optimization & power system models
- ›› Julia, Python, and tools for energy analytics
- ›› Simulation, forecasting, and uncertainty analysis

Primary Channel: Email

- › **Official Email:**
luislopezdiaz@cmail.carleton.ca
- › **Fastest Response:** Email is the quickest way to reach me, I typically reply promptly.
- › **Subject Tag:** Always include [MAAE 2300] in the subject line for automatic filtering.

Office Hours: By Appointment

- › No Fixed Drop-in Hours
- › **When Needed:** If a question requires deeper discussion beyond email, request a meeting.
- › **Format:** Available **in-person** on campus or **virtually** via MS Teams.

Icebreaker Activity

- 1 Meet Your TA
- 2 Icebreaker Activity

- 3 Tutorial Methodology
- 4 Keys to Success

Icebreaker Dynamic: Breaking the Ice!



The 15-Second Student Check-In

Around the room, share with us:

1

Your Name
(or preferred name)

2

Your Program
(Mechanical,
Aero, Civil, etc.)

3

Favorite Pick
(Movie, TV se-
ries, or hobby)



Don't overthink it — whatever show or movie you watched most recently counts!

Tutorial Methodology

- 1 Meet Your TA
- 2 Icebreaker Activity

- 3 Tutorial Methodology
- 4 Keys to Success

The 6-Step Engineering Problem-Solving Framework

➔ Step 1: Sketch & Control Volume

Draw the system, select control volume (CV) or fluid element, and fix coordinate axes (x, y, z).

➔ Step 2: Given & Target Parameters

Extract numerical and symbolic data with correct SI units; identify explicitly what must be solved.

➔ Step 3: Engineering Assumptions

State assumptions (*steady, incompressible, inviscid, 1D flow*); justify dropping individual terms.

➔ Step 4: Governing Conservation Laws

Select fundamental laws (Continuity, Momentum, Energy, Hydrostatic relation, Navier-Stokes).

➔ Step 5: Symbolic Derivation First

Complete all algebraic simplifications in symbolic form **before** plugging in numbers!

✓ Step 6: Calculate & Sanity Check

Substitute numbers with units; perform order-of-magnitude and physical sanity checks.

Keys to Success

- 1 Meet Your TA
- 2 Icebreaker Activity

- 3 Tutorial Methodology
- 4 Keys to Success

Practical Keys to Acing Fluid Mechanics I

💡 Study & Exam Habits

- **Understand, Don't Memorize:** Fluid mechanics has few fundamental laws; all formulas are simply conservation of mass, momentum, or energy applied to a CV.
- **Units are Your Radar:** Always track units (N s m^{-2} , Pa, m s^{-1}). If units don't balance, the equation is wrong!
- **Draw First, Write Later:** A clear, labeled diagram solves 50% of any fluid mechanics problem.

⚠ Classic Pitfalls to Avoid

- **Gauge vs. Absolute Pressure:** Forgetting that ideal gas law $P = \rho RT$ and thermodynamic equations require **absolute pressure** ($P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}}$).
- **Premature Calculation:** Rounding numbers at step 1 instead of simplifying algebraically.
- **Skipping Assumptions:** Applying Bernoulli equation without verifying irrotational/inviscid/steady conditions.



Let's Have a Great Term Together!